Deep Reinforcement Learning

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Lecture 1: Introduction and Course Overview

1.1 Learning to make decisions

Reinforcement learning (RL) studies how to learn decisions and control from experience. An agent selects actions, observes their consequences, and receives rewards that describe aspects of success. Its choices influence what it will encounter next, so the data collected by a decision maker depends on its own behavior.

This feedback makes the problem different from standard supervised learning, where training examples are often assumed independent and identically distributed and a target output is supplied for each input. In RL the correct action is generally unknown: a reward evaluates an outcome but need not say which earlier action caused it or what should have been done instead. A decision can have consequences well after it is made.

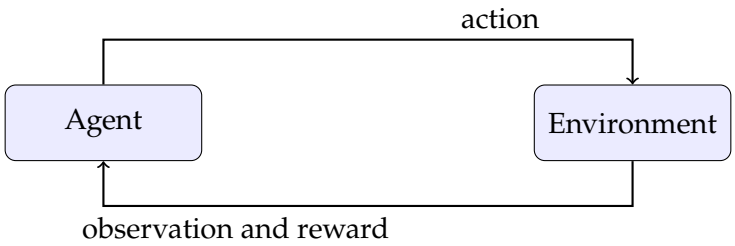


Figure 1: Actions change the experience from which the agent learns.

The same interaction has different meanings in different settings. For a robot, actions may be motor torques, observations camera images, and reward a measure of task success such as running speed. For inventory management, actions are purchases, observations include stock levels, and profit can provide reward. The agent must learn how its actions affect future situations, not merely predict a label for the current input.

1.2 Learning and optimization

Data-driven learning extracts patterns about the world from experience; optimization uses computation to choose behavior that serves a goal. The two are complementary. Learning from examples alone can reproduce patterns in the available data, while optimizing in a simulator or a hand-built model can find solutions that exploit inaccuracies in that model. Combining learning and optimization aims to use real experience while discovering behaviors beyond those explicitly demonstrated. AlphaGo's surprising move 37 against Lee Sedol illustrates the possibility of finding a successful decision that people had not anticipated.

Deep networks offer a way to learn from large, complex datasets and process rich inputs, while RL provides a framework for optimizing decisions. Applications span games, physical manipulation and locomotion, traffic, language models, image generation, and chip design. A successful demonstration in one setting does not by itself solve the problem of reliable control in the real world.

1.3 Beyond a given reward

A reward is only one form of supervision for decision making. A demonstration can be used for IMITATION LEARNING, which directly learns to reproduce observed behavior. Alternatively, INVERSE REINFORCEMENT LEARNING tries to infer a reward or preference that explains the behavior, allowing an agent to pursue that inferred objective in a different situation. These are different uses of the same example: copying an action is not the same as inferring its purpose.

Observations can also train PREDICTIVE MODELS of how the world changes. Such a model can support action selection by comparing predicted consequences of possible actions. Knowledge from other tasks or pretrained models can help with a new task; TRANSFER LEARNING reuses that knowledge, while META-LEARNING aims to improve the process of learning new tasks.

1.4 Open challenges

Deep RL can require large amounts of experience, whereas people often learn quickly and reuse prior knowledge. Important open problems include making learning more data efficient, transferring skills across tasks, specifying or learning suitable rewards, and determining how prediction should support decision making. Rich perception and complex action must work together in an agent that adapts through interaction with its environment.

Source. Sergey Levine, Deep Reinforcement Learning, Fall 2023, Lecture 1.